

Depending on the active compound's specific method of action, sunscreens are broadly classified into two groups: organic and inorganic. Organic sunscreens typically contain aromatic compounds that absorb ultraviolet (UV) radiation. The absorbed energy from the UV radiation causes these compounds to enter electronically excited states that then relax back to the ground state and harmlessly re-emit the absorbed radiation as heat. Alternatively, inorganic sunscreens contain metallic nanoparticles that serve to reflect, scatter, and absorb incident radiation. The mechanism for UV protection of each group is shown in Figure 1.

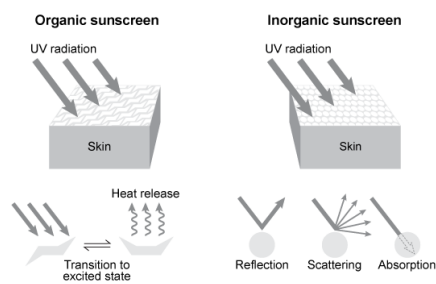


Figure 1 Method of action for organic and inorganic sunscreens

The effectiveness of sunscreens is traditionally evaluated on their ability to prevent reddening of the skin, as indicated by their sun protection factor (SPF). However, studies have shown that additional deleterious effects, such as the suppression of T cell-mediated immune responses, can occur at lower UV doses than erythema, indicating the need for new metrics to judge sunscreen efficacy. For this reason, researchers conducted a study to compare the protective ability of six sunscreens against photo-isomerization of the photoreceptor *trans*-urocanic acid (UCA) to its *cis* form, a process believed to be linked to UV-induced immunosuppression. Table 1 lists the sunscreens tested and their relevant characteristics.

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In vivo experiments were conducted on two groups of human volunteers. Participants in the acute exposure group received a single UV dose of 100 mJ/cm² whereas participants in the chronic exposure group received the same dosage once daily over four consecutive days. Each sunscreen was applied to a different spot on the volunteers' back prior to irradiation. A portion of skin without sunscreen was also irradiated to serve as a control. After irradiation, the sunscreens were removed and the skin was wiped with filter papers. Filters were washed with potassium hydroxide, and the aqueous solution was analyzed for UCA using high-performance liquid chromatography. Figure 2 shows the mean percentages of *cis*-UCA formed upon irradiation for each exposure type and sunscreen tested.

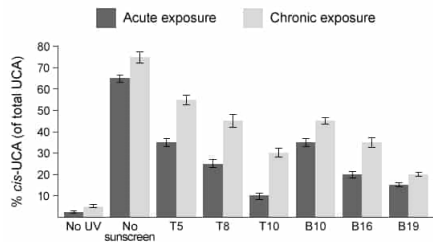


Figure 2 Percentage of UCA in *cis* configuration upon UV exposure with various sunscreens

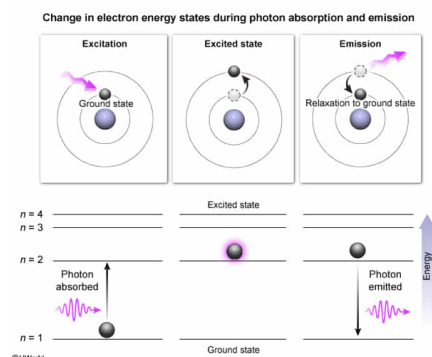
Which of the following statements is true concerning the sunscreens discussed in the passage?

- ✓ ☒ A. Organic sunscreens' method of protection involves relaxation to the ground state. [73%]
- ☐ B. Sunscreens with titanium dioxide offer the greatest range of protection. [4%]
- ☐ C. For chronic exposure to UV radiation, titanium dioxide provides better protection from photo-isomerization of *trans*-UCA to *cis*-UCA than does octisalate at an equal concentration. [13%]
- ☐ D. A sunscreen's SPF can be used to evaluate its efficacy in protection from UV-induced immunosuppression. [8%]

Correct

74% answered correctly

Explanation:



Orbitals are areas in space around the nucleus of an atom in which electrons may be found. Due to the quantized nature of electrons, **orbitals** and the electrons that occupy them exist only in specific sets of **energy levels**. Electrons can **transition** from one allowed **energy level** to another by absorbing or emitting energy equal to the **difference between two energy levels**. To enter an **excited state**, electrons **absorb energy**; after transitioning to an excited state, electrons must **relax** back to their **ground state**, the lowest available energy level, through the **emission** of energy.

As stated in the passage, organic sunscreens are aromatic compounds that protect skin by absorbing ultraviolet (UV) radiation. The electrons that absorb energy during this process enter electronically excited states and must subsequently relax back down to their ground state. This occurs by the emission of the excitation energy as heat.

(Choice B) As shown in Table 1, sunscreens with titanium dioxide (T5, T8, and T10) offer protection from wavelengths of 280–340 nm, which is less than the range of protection offered by sunscreen B19 at 250–370 nm.

(Choice C) Table 1 shows that sunscreens T8 and B10 have equal concentrations of titanium dioxide and octisalate, respectively. Figure 2 shows that these two sunscreens provide similar protection from photo-isomerization.

(Choice D) The passage indicates that SPF is useful in evaluating a sunscreen's ability to protect from erythema of the skin. Because photo-isomerization (and resulting immunosuppression) occurs at lower doses of UV than erythema, the SPF value may not correlate with the sunscreen's ability to protect from photo-induced immunosuppression.

Educational objective:

Electrons occupy orbitals around a nucleus, each with a specific energy. Electrons can jump between energy levels by absorbing or emitting energy exactly equal to the difference in energy levels. After excitation, electrons must relax back to their ground state.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400314
System:
Atomic Theory and Chemical Composition

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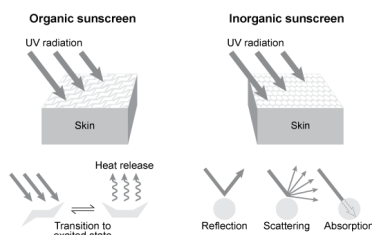


Figure 1 Method of action for organic and inorganic sunscreens

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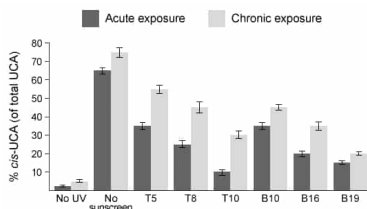
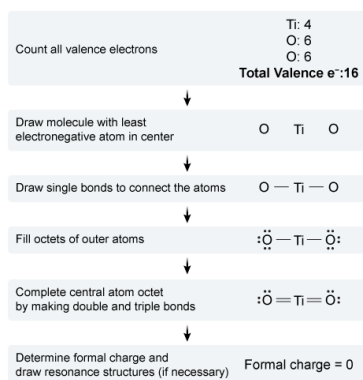


Figure 2 Percentage of UCA in *cis* configuration upon UV exposure with various sunscreens

Which of the following is the correct Lewis dot structure for the common inorganic sunscreen compound titanium dioxide (TiO₂) given that titanium has four valence electrons?

- ☐ A. $\text{:}\ddot{\text{O}}\text{--Ti--}\ddot{\text{O}}\text{:}$ [13%]
- ☐ B. $\text{:}\ddot{\text{O}}\text{=Ti=}\ddot{\text{O}}\text{:}$ [11%]
- ☐ C. $\text{:}\ddot{\text{O}}\text{--}\ddot{\text{T}}\text{i=}\ddot{\text{O}}\text{:}$ [1%]
- ☒ D. $\text{:}\ddot{\text{O}}\text{=Ti=}\ddot{\text{O}}\text{:}$ [74%]

Explanation:

Lewis dot structures are used to represent the covalent bonding between the atoms of a molecule. **Single bonds**, which consist of 2 electrons, are represented as a line between element symbols, and unshared **nonbonding electrons** (ie, lone pairs) are represented as dots.

Lewis dot structures are constructed using the following steps:

1. Determine the number of **valence electrons**, which is the number of electrons in the atom's outer shell, for **every atom in the molecule**. Titanium has 4 and each oxygen has 6, for a total of 16 in titanium dioxide (TiO₂).
2. Draw the structure of the compound with the atom that has the lowest **electronegativity** positioned in the **center**, and then connect the atoms with single bonds. Ti is the least electronegative atom in TiO₂.
3. Add dots (representing electrons) to the atoms surrounding the central atom so they have **complete octets** (be aware of atoms with **exceptions to the octet rule**). Remember that each bond represents 2 electrons, and the total number of electrons per atom should be 8.
4. Place any remaining electrons on the central atom. In the case of TiO₂, all 16 electrons are already accounted for (ie, no electrons are left to place). Oxygen is a **divalent compound** (ie, typically forms two bonds). Consequently, 2 electrons from the lone pairs on each oxygen are used to form an additional bond with titanium on each side, making two double bonds and completing titanium's octet.
5. Ensure the resulting Lewis dot structure is the most accurate representation by calculating the **formal charge**, which should be equal to the final charge of the compound (or zero for neutral molecules such as TiO₂ that have no net charge).

(Choice A) Titanium should have 8 electrons for a full octet, but only 4 are shown.

(Choice B) The oxygen atoms each have 10 electrons, which is not allowed according to the octet rule.

(Choice C) A total of 18 valence electrons are shown, but TiO₂ should only have 16 electrons. In addition, TiO₂ is a neutral molecule, and therefore the formal charge should be zero.

Educational objective:

Lewis dot structures represent the distribution of bonds and valence electrons between atoms in a molecule, with the least electronegative atom in the center. A correct Lewis dot structure shows all valence electrons and complete octets for each atom and yields the correct formal charge.

Time spent: 0 sec
Subject:
 General Chemistry

QID: 400315
System:
 Atomic Theory and Chemical Composition

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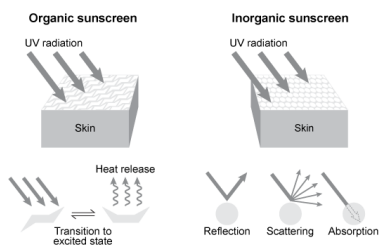


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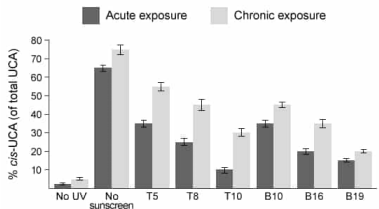


Figure 2 Percentage of UCA in *cis* configuration upon UV exposure with various sunscreens

What is the molarity of cinoxate in 100 mL of sunscreen B16? (Note: The molecular weight of cinoxate is 250 g/mol, and the density of B16 is 1.0 g/mL.)

- ☐ A. 0.0010 M [15%]
- ☒ B. 1.0 M [43%]
- ☐ C. 4.0 M [38%]
- ☐ D. 6,250 M [2%]

Correct
42% answered correctly

Explanation:

Calculate mass of solution	$m_{\text{solution}} = \rho_{\text{solution}} \cdot V_{\text{solution}} = 100 \text{ g}$
Calculate mass of cinoxate	$m_{\text{cinoxate}} = \frac{(m_{\text{solution}} \cdot \% \text{ cinoxate})}{100} = 25 \text{ g}$
Calculate moles of cinoxate	$\text{mol}_{\text{cinoxate}} = \frac{m_{\text{cinoxate}}}{\text{MW}} = 0.10 \text{ mol}$
Calculate molarity of cinoxate	$M_{\text{cinoxate}} = \frac{\text{mol}_{\text{cinoxate}}}{L_{\text{solution}}} = 1.0 \text{ M}$

The **molarity** (ie, **molar concentration**) of any solution is calculated as the moles of solute divided by the total volume of solution in liters. To find the moles of a compound, its mass must first be determined. The mass can then be converted to moles using the **molar mass** (ie, molecular weight) of the compound.

The total mass m_{solution} of the sunscreen B16 sample can be found by multiplying its **density** ρ_{solution} by its volume V_{solution} .

$$m_{\text{solution}} = \rho_{\text{solution}} \cdot V_{\text{solution}}$$

$$m_{\text{solution}} = \left(1.0 \frac{\text{g}}{\text{mL}}\right) \cdot (100 \text{ mL}) = 100 \text{ g}$$

Table 1 indicates that sunscreen B16 is **25% cinoxate by mass**. As such, the mass of cinoxate present in the total mass of the sunscreen sample is given by:

$$m_{\text{cinoxate}} = \frac{(m_{\text{solution}} \cdot \% \text{ cinoxate})}{100}$$

$$m_{\text{cinoxate}} = \frac{(100 \text{ g} \cdot (25\%))}{100} = (100 \text{ g}) \cdot (0.25) = 25 \text{ g}$$

The number of moles of cinoxate is found by dividing the mass of cinoxate by its **molecular weight** (MW).

$$25 \text{ g cinoxate} \times \frac{1 \text{ mol cinoxate}}{250 \text{ g cinoxate}} = 0.10 \text{ mol cinoxate}$$

Dividing the moles of cinoxate in the sample by the 100 mL sample volume (converted to liters) gives the **molarity** of cinoxate in the sample.

$$\text{Molarity} = \frac{\text{moles solute}}{\text{volume of solution(L)}} = \frac{0.10 \text{ mol cinoxate}}{0.100 \text{ L}} = 1.0 \text{ M}$$

(Choice A) A value of 0.0010 M (with incorrect units) results from not converting 100 mL to 0.100 L during the calculation. Molarity is defined as moles per *liter* (mol/L).

(Choice C) A concentration of 4.0 M is obtained by incorrectly taking the entire 100 g of the B16 sample as cinoxate. Table 1 indicates that sunscreen B16 is only 25% cinoxate by mass.

(Choice D) A value of 6,250 M (with incorrect units) results from multiplying the mass of cinoxate by its molar mass (g/mol) instead of dividing by its molar mass.

Educational objective:

The density, percent mass, and molecular weight of a substance can be used to calculate the molar concentration (molarity) of a solution. The units must be consistent when carrying out these stoichiometric calculations.

Time spent: 0 sec
Subject:
 General Chemistry

QID: 400316
System:
 Atomic Theory and Chemical Composition

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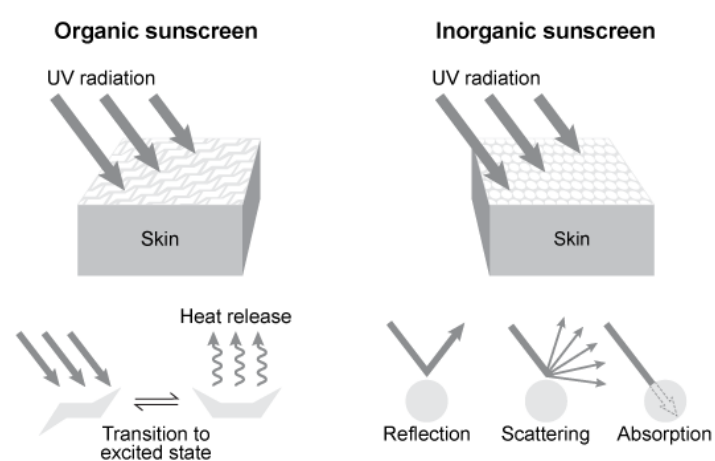


Figure 1 Method of action for organic and inorganic sunscreens

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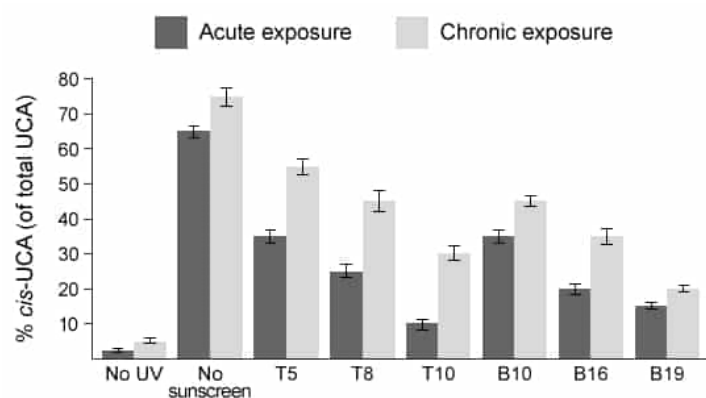


Figure 2 Percentage of UCA in *cis* configuration upon UV exposure with various sunscreens

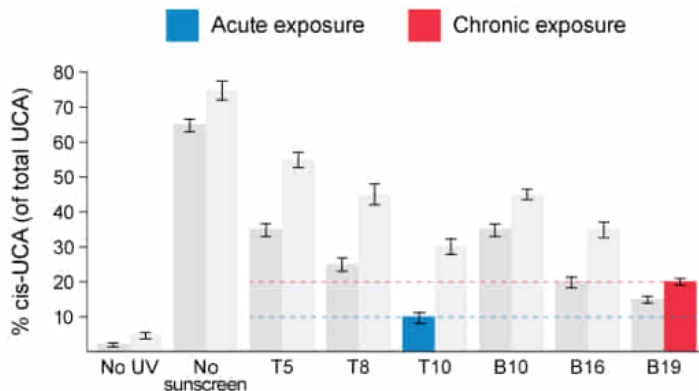
Based on the results from the experiment shown in Figure 2, which of the following conclusions can be drawn?

- ☐ A. Acute exposure to UV radiation leads to greater immunosuppression than chronic exposure. [3%]
- ☐ B. Without sunscreen, UV radiation causes one in three UCA molecules to become *cis* isomers. [4%]
- ✓ ☒ C. T10 and B19 offer the best protection against acute and chronic UV exposure, respectively. [88%]
- ☐ D. T5 offers the best overall protection from UV radiation. [3%]

Correct

87% answered correctly

Explanation:



According to the passage, photo-isomerization of *trans*-UCA to *cis*-UCA is linked to UV-induced immunosuppression. Therefore, a sunscreen with a **smaller percentage** of *cis*-UCA provides **better protection** than a sunscreen that permits a greater percentage of *cis*-UCA formation.

Figure 2 shows the mean percentages of *cis*-UCA formed during acute and chronic exposure for each sunscreen tested. The graph indicates that under acute exposure, skin treated with T10 sunscreen yields the least *cis*-UCA formation whereas under chronic exposure, skin treated with B19 sunscreen yields the least *cis*-UCA formation. Therefore, sunscreens T10 and B19 offer the best protection against acute and

chronic UV exposure, respectively.

(Choice A) Acute exposure leads to a smaller amount of *cis*-UCA formation than chronic exposure for all sunscreens tested in Figure 2. A smaller amount of *cis*-UCA indicates less immunosuppression.

(Choice B) Exposure without sunscreen leads to 65% *cis*-UCA for acute exposure and 75% *cis*-UCA for chronic exposure. Therefore, UV radiation causes approximately two in three (acute exposure) and three in four (chronic exposure) UCA molecules to become *cis* isomers.

(Choice D) T5 provides the least protection from UV radiation because it leads to the highest percentage of *cis*-UCA formation for chronic exposure, which indicates a greater instance of UV-induced immunosuppression.

Educational objective:

Careful analysis of data and what it implies is important to form conclusions from experimental results. A correct understanding of what the data indicate is essential for drawing correct conclusions.

Time spent:0 sec

Subject:
General Chemistry

QID:400317

System:
Atomic Theory and Chemical Composition

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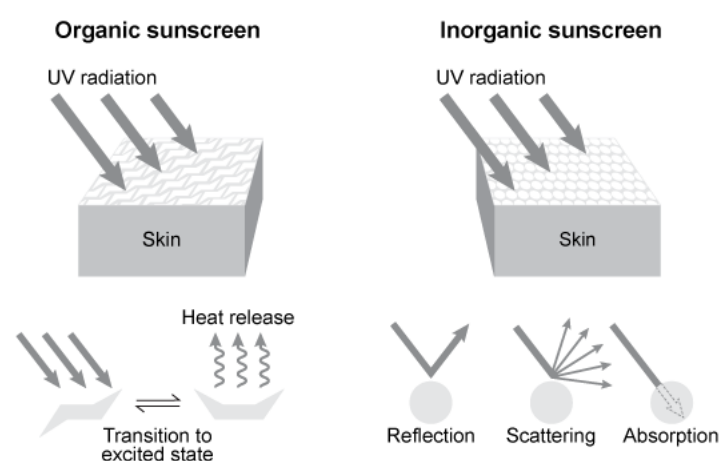


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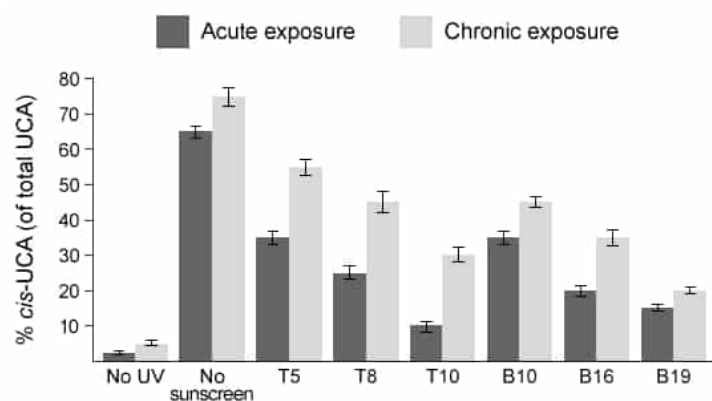


Figure 2 Percentage of UCA in *cis* configuration upon UV exposure with various sunscreens

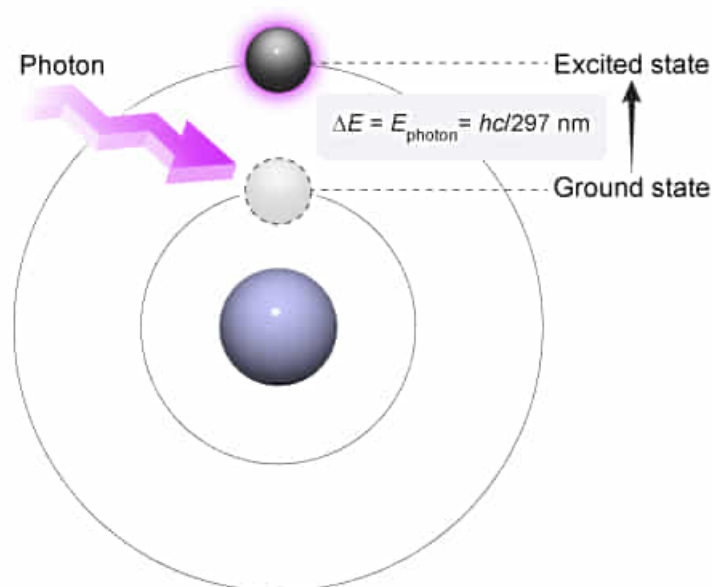
Which of the following is true concerning the energy difference ΔE between the ground state and the photo-excited state that occurs *most often* for the active compound in sunscreen B10?

- ☐ A. $\Delta E = hc \times 297 \text{ nm}$ [4%]
- ☐ B. $\Delta E = hc \times (250 \text{ nm} - 300 \text{ nm})$ [20%]
- ☐ C. $\Delta E = \frac{hc}{250 \text{ nm}} + \frac{hc}{300 \text{ nm}}$ [18%]
- ✓ ☒ D. $\Delta E = \frac{hc}{297 \text{ nm}}$ [56%]

Correct

55% answered correctly

Explanation:



The **energy of a photon** E_{photon} is determined according to:

$$E_{\text{photon}} = \frac{hc}{\lambda}$$

where h is Planck's constant, c is the speed of light, and λ is the **wavelength** of the photon.

Table 1 shows that the peak protection of sunscreen B10 is at 297 nm (ie, the active compound most commonly absorbs photons of that wavelength). Because an atom or molecule can only **absorb a photon** with an **energy exactly equal** to the **difference between two energy levels**, the **difference in energy** ΔE between the **ground state** and the most commonly occurring **excited state** for the active compound in sunscreen B10 is:

$$\Delta E = E_{\text{photon}} = \frac{hc}{297 \text{ nm}}$$

(Choice A) The energy of a photon is **inversely proportional** to its wavelength, so hc must be divided (not multiplied) by 297 nm.

(Choice B) Subtracting the energy of a 300-nm photon from that of a 250-nm photon would give the energy difference between the photons of those wavelengths, but this value is not the energy difference between the ground state and excited state of a molecule.

(Choice C) Although B10 can absorb photons with wavelengths in the range of 250 nm to 300 nm, each absorption event occurs for only one photon at a time. The molecule would have to simultaneously absorb a photon at 250 nm and a photon at 300 nm for this answer to be correct.

Educational objective:

The energy of a photon is given by $E_{\text{photon}} = \frac{hc}{\lambda}$. For an electron in an atom or molecule to transition from its ground state to an excited state, the atom or molecule must absorb a single photon with an energy exactly equal to the energy difference between the two states.

Time spent:0 sec

Subject:
General Chemistry

QID:400318

System:
Atomic Theory and Chemical Composition

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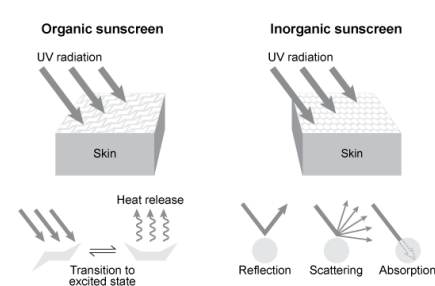


Figure 1 Method of action for organic and inorganic sunscreens

The effectiveness of sunscreens is traditionally evaluated on their ability to prevent reddening of the skin, as indicated by their sun protection factor (SPF). However, studies have shown that additional deleterious effects, such as the suppression of T cell-mediated immune responses, can occur at lower UV doses than erythema, indicating the need for new metrics to judge sunscreen efficacy. For this reason, researchers conducted a study to compare the protective ability of six sunscreens against photo-isomerization of the photoreceptor *trans*-urocanic acid (UCA) to its *cis* form, a process believed to be linked to UV-induced immunosuppression. Table 1 lists the sunscreens tested and their relevant characteristics.

Table 1 Results from Tested Sunscreens

Sunscreen	Active compound(s) (mass %)	Range of protection (nm)	Greatest protection (nm)
T5	10% Titanium dioxide	280–340	300
T8	20% Titanium dioxide	280–340	300
T10	30% Titanium dioxide	280–340	300
B10	20% Octisalate	250–300	297
B16	25% Cinoxate	260–318	280
B19	8% Butyl methoxydibenzoylmethane,	250–370	310
	12% 4-methylbezylidene		

In vivo experiments were conducted on two groups of human volunteers. Participants in the acute exposure group received a single UV dose of 100 mJ/cm² whereas participants in the chronic exposure group received the same dosage once daily over four consecutive days. Each sunscreen was applied to a different spot on the volunteers' back prior to irradiation. A portion of skin without sunscreen was also irradiated to serve as a control. After irradiation, the sunscreens were removed and the skin was wiped with filter papers. Filters were washed with potassium hydroxide, and the aqueous solution was analyzed for UCA using high-performance liquid chromatography. Figure 2 shows the mean percentages of *cis*-UCA formed upon irradiation for each exposure type and sunscreen tested.

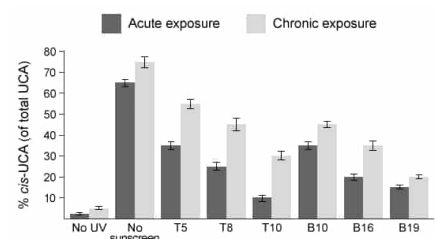


Figure 2 Percentage of UCA in *cis* configuration upon UV exposure with various sunscreens

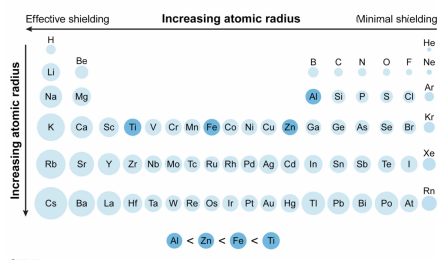
Zinc, titanium, iron, and aluminum are all ingredients found in inorganic sunscreens. Which of the following choices correctly lists the atoms in order of increasing atomic radius?

- ✓ ☒ A. $\text{Al} < \text{Zn} < \text{Fe} < \text{Ti}$ [74%]
☐ B. $\text{Zn} < \text{Fe} < \text{Ti} < \text{Al}$ [2%]
☐ C. $\text{Ti} < \text{Fe} < \text{Zn} < \text{Al}$ [10%]
☐ D. $\text{Al} < \text{Ti} < \text{Fe} < \text{Zn}$ [12%]

Correct

74% answered correctly

Explanation:



An **atomic radius** is defined as the distance from the nucleus to the outermost electron shell of a neutral atom that retains all its electrons (ie, the atom has not been converted into an ion). In contrast, an **ionic radius** measures the radius of an atom that has gained or lost electrons (often an entire **electron shell**). As such, atomic and ionic radius measurements are distinct and must not be treated equivalently.

The **periodic table** has the following trends for the **atomic radius** of atoms:

- Within a **group** (column), the atomic radius **increases** from **top to bottom** because each successive atom in the group has an additional electron shell. The outer **valence electrons** are shielded from the pull of the protons by the inner **core electrons**. The valence electrons experience a decreased attractive force toward the nucleus (ie, **effective nuclear charge**, Z_{eff}), so they orbit farther from the nucleus.
- Within each **period** (row), the atomic radius gradually **increases** from **right to left**. With each step to the right, an electron and a proton are added. The charge in the nucleus increases with each additional proton, causing a **gradual drawing in** of the electron cloud. Because the number of electron shells is constant within a period, each new electron is added to the same shell and cannot effectively shield other electrons from the nucleus.

Because the addition of shells has a strong effect on **shielding**, changes in atomic radius **down a group** are **more pronounced** than for those across a period.

Aluminum has the smallest atomic radius because it is located in the highest row and farthest to the right in the periodic table. Zinc, iron, and titanium are all in the same period, so the trend from right to left is $\text{Al} < \text{Zn} < \text{Fe} < \text{Ti}$.

(Choice B) Aluminum has the smallest atomic radius rather than the largest.

(Choice C) Both trends have been reversed. Atomic radius decreases to the right across a period and increases down a group.

(Choice D) Although aluminum has the smallest atomic radius, the trend for the remaining atoms is reversed.

Educational objective:

Atomic radii are governed by the distribution of electrons surrounding the atom's nucleus. They quickly increase from top to bottom in a group and gradually decrease from left to right across a period on the periodic table.

Time spent: 0 sec
 Subject:
 General Chemistry

QID: 400319
 System:
 Atomic Theory and Chemical Composition